PYTHON ACTIVITY BASED LEARNING

Problem-3:

Student Email and Identifier Validation: Validate student email-like strings and registration identifiers against supplied formatting rules, reporting malformed separators, invalid character groups, and missing components.

Objective

To develop a Python program that accepts student email addresses and registration IDs as user input, validates their formats using predefined rules, and displays whether each input is valid or invalid along with the corresponding error message.

Algorithm

1. Start the program.
2. Read the number of students from the user.
3. For each student, accept an email address.
4. Check whether the email contains exactly one @ symbol.
5. Validate the username, domain, dot separators, and allowed characters.
6. Display the email as valid or invalid with the appropriate error message.
7. Accept the registration ID for each student.
8. Check whether the registration ID contains exactly six characters, starts with BT, and ends with four digits.
9. Display the registration ID as valid or invalid with the appropriate error message.
10. Repeat until all students are processed.
11. Stop the program.

Pseudocode

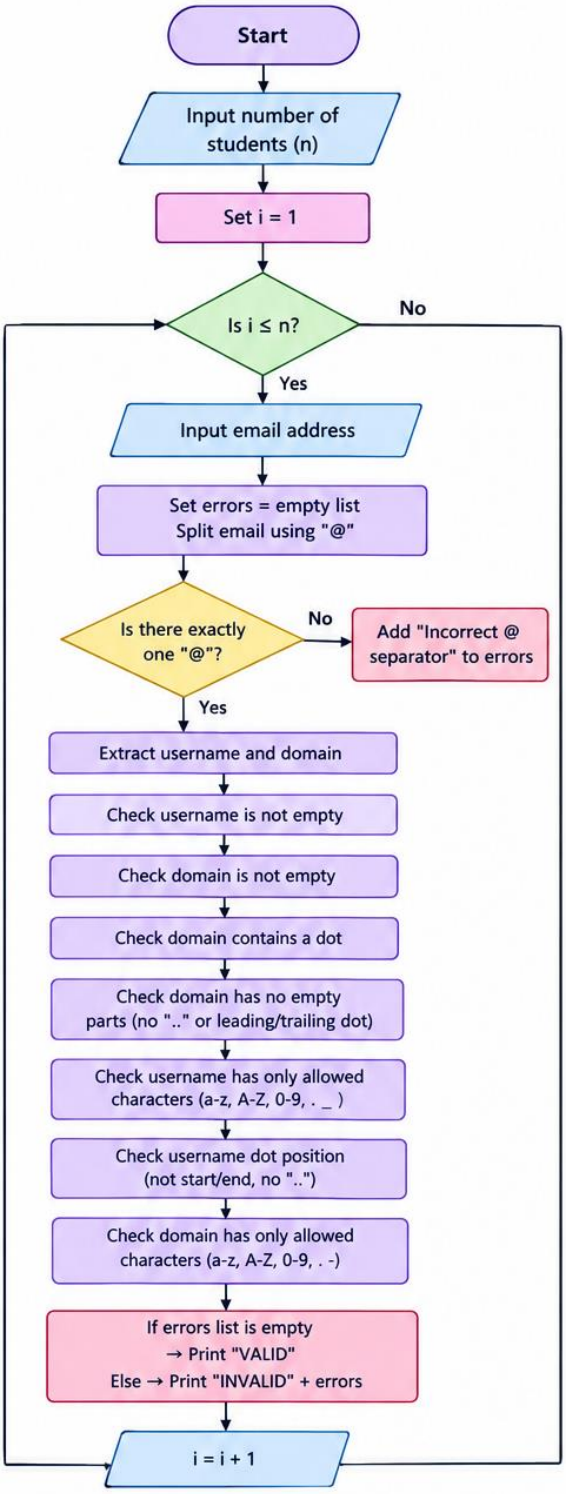
BEGIN

```
INPUT number_of_students
PRINT "STUDENT EMAIL VALIDATION"
FOR each student from 1 to number_of_students DO
    INPUT email
    SET errors = empty list
    SPLIT email using "@"
    IF email contains not exactly one "@" THEN
        ADD "Incorrect @ separator" to errors
    ELSE
        EXTRACT username and domain
        CHECK username is not empty
        CHECK domain is not empty
        CHECK domain contains a dot
        CHECK domain has no empty dot-separated parts
        CHECK username contains only allowed characters
        CHECK username has valid dot positions
        CHECK domain contains only allowed characters
    END IF
    IF errors is empty THEN
        PRINT email, "VALID"
    ELSE
        PRINT email, "INVALID", errors
    END IF
END FOR
PRINT "REGISTRATION ID VALIDATION"
FOR each student from 1 to number_of_students DO
```

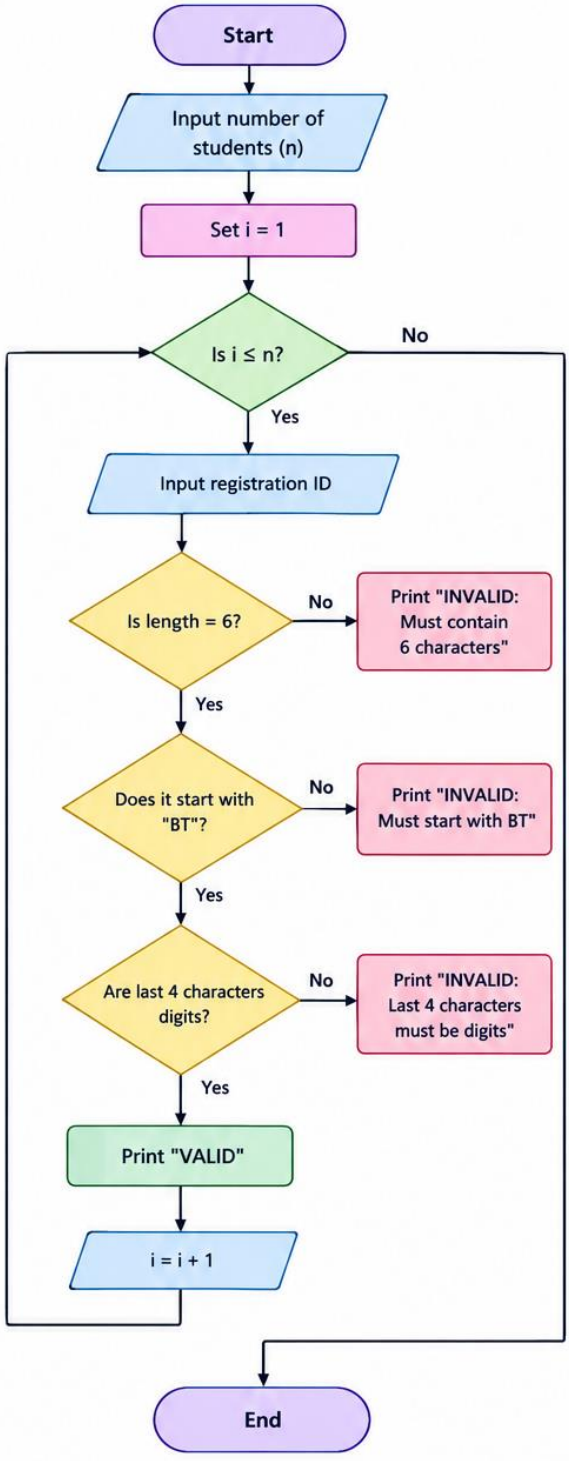
```
    INPUT registration_id
    IF length of registration_id is not 6 THEN
        PRINT "INVALID: Must contain 6 characters"
    ELSE IF registration_id does not start with "BT" THEN
        PRINT "INVALID: Must start with BT"
    ELSE IF last four characters are not digits THEN
        PRINT "INVALID: Last 4 characters must be digits"
    ELSE
        PRINT registration_id, "VALID"
    END IF
END FOR
STOP
```

Flowchart:

Email Validation Flowchart



Registration ID Validation Flowchart



Objective:

To develop a Python program that accepts student email addresses and registration IDs as user input, validates their formats using predefined rules, and displays whether each input is valid or invalid along with the corresponding error message.

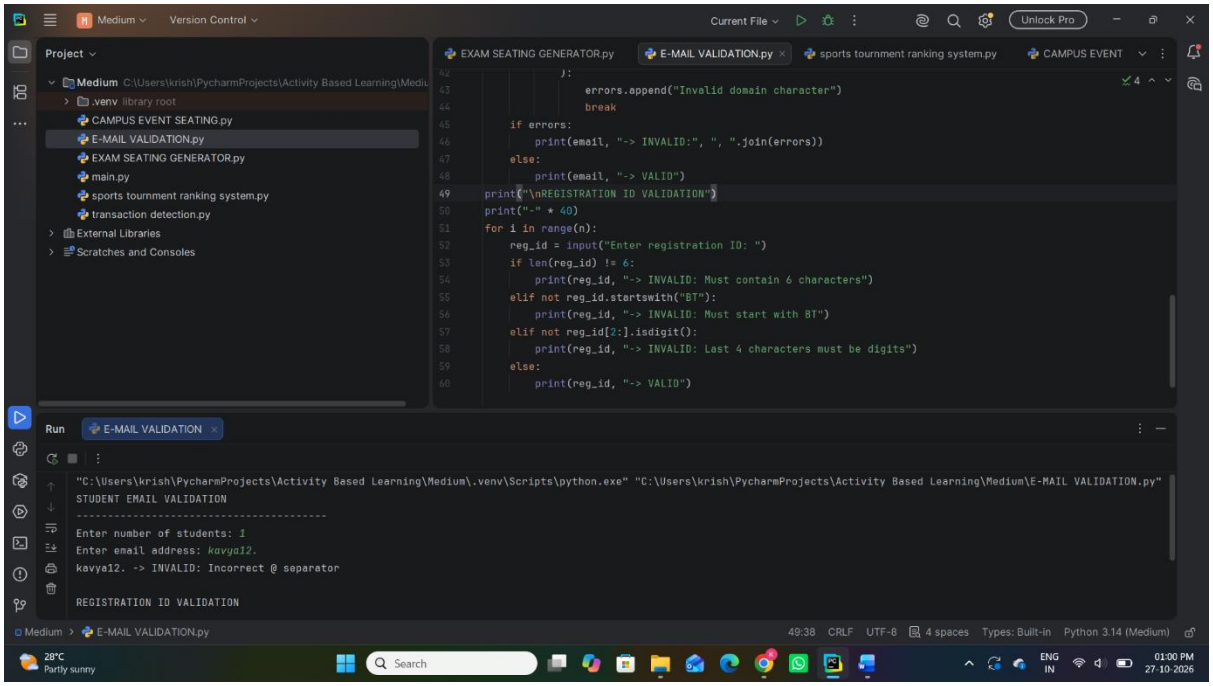
Python Program:

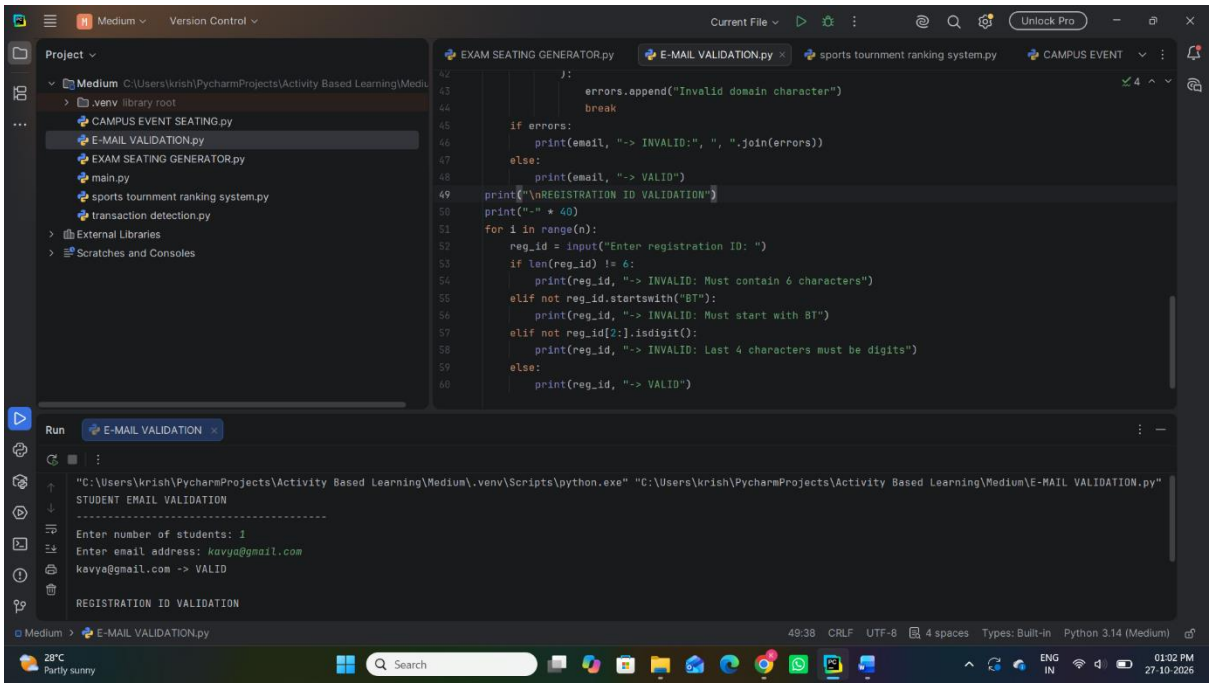
```
print("STUDENT EMAIL VALIDATION")
print("-" * 40)

n = int(input("Enter number of students: "))
for i in range(n):
    email = input("Enter email address: ")
    errors = []
    parts = email.split("@")
    if email.count("@") != 1:
        errors.append("Incorrect @ separator")
    else:
        username = parts[0]
        domain = parts[1]
        if username == "":
            errors.append("Missing username")
        if domain == "":
            errors.append("Missing domain")
        elif "." not in domain:
            errors.append("Missing domain extension")
        else:
            domain_parts = domain.split(".")
            if any(part == "" for part in domain_parts):
                errors.append("Malformed dot separator")
        allowed = (
            "abcdefghijklmnopqrstuvwxyz"
            "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
            "0123456789._"
        )
        for ch in username:
            if ch not in allowed:
                errors.append("Invalid username character")
                break
        if username.startswith(".") or username.endswith("."):
            errors.append("Invalid dot position")
        if ".." in username:
            errors.append("Repeated dots in username")
```

```
        for ch in domain:
            if ch not in (
                "abcdefghijklmnopqrstuvwxyz"
                "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
                "0123456789.-"
            ):
                errors.append("Invalid domain character")
                break
    if errors:
        print(email, "-> INVALID:", ", ".join(errors))
    else:
        print(email, "-> VALID")
print("\nREGISTRATION ID VALIDATION")
print("-" * 40)
for i in range(n):
    reg_id = input("Enter registration ID: ")
    if len(reg_id) != 6:
        print(reg_id, "-> INVALID: Must contain 6 characters")
    elif not reg_id.startswith("BT"):
        print(reg_id, "-> INVALID: Must start with BT")
    elif not reg_id[2:].isdigit():
        print(reg_id, "-> INVALID: Last 4 characters must be digits")
    else:
        print(reg_id, "-> VALID")
```

Output Screen Shots:


Medium
Version Control
Current File
Unlock Pro
Project
Medium C:\Users\krish\PycharmProjects\Activity Based Learning\Medi
.venv library root
CAMPUS EVENT SEATING.py
E-MAIL VALIDATION.py
EXAM SEATING GENERATOR.py
main.py
sports tournment ranking system.py
transaction detection.py
External Libraries
Scratches and Consoles
EXAM SEATING GENERATOR.py
E-MAIL VALIDATION.py
sports tournment ranking system.py
CAMPUS EVENT
):
errors.append("Invalid domain character")
break
if errors:
print(email, "-> INVALID:", ", ".join(errors))
else:
print(email, "-> VALID")
print("\nREGISTRATION ID VALIDATION")
print("-" * 40)
for i in range(n):
reg_id = input("Enter registration ID: ")
if len(reg_id) != 6:
print(reg_id, "-> INVALID: Must contain 6 characters")
elif not reg_id.startswith("BT"):
print(reg_id, "-> INVALID: Must start with BT")
elif not reg_id[2:].isdigit():
print(reg_id, "-> INVALID: Last 4 characters must be digits")
else:
print(reg_id, "-> VALID")
Run
E-MAIL VALIDATION
"C:\Users\krish\PycharmProjects\Activity Based Learning\Medium\.venv\Scripts\python.exe" "C:\Users\krish\PycharmProjects\Activity Based Learning\Medium\E-MAIL VALIDATION.py"
STUDENT EMAIL VALIDATION
Enter number of students: 1
Enter email address: kavya12.
kavya12. -> INVALID: Incorrect @ separator
REGISTRATION ID VALIDATION
Medium > E-MAIL VALIDATION.py
49:38 CRLF UTF-8 4 spaces Types: Built-in Python 3.14 (Medium)
28°C
Partly sunny
Search
ENG
IN
01:00 PM
27-10-2026


Medium
Version Control
Current File
Unlock Pro
Project
Medium C:\Users\krish\PycharmProjects\Activity Based Learning\Medi
.venv library root
CAMPUS EVENT SEATING.py
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transaction detection.py
External Libraries
Scratches and Consoles
EXAM SEATING GENERATOR.py
E-MAIL VALIDATION.py
sports tournment ranking system.py
CAMPUS EVENT
):
errors.append("Invalid domain character")
break
if errors:
print(email, "-> INVALID:", ", ".join(errors))
else:
print(email, "-> VALID")
print("\nREGISTRATION ID VALIDATION")
print("-" * 40)
for i in range(n):
reg_id = input("Enter registration ID: ")
if len(reg_id) != 6:
print(reg_id, "-> INVALID: Must contain 6 characters")
elif not reg_id.startswith("BT"):
print(reg_id, "-> INVALID: Must start with BT")
elif not reg_id[2:].isdigit():
print(reg_id, "-> INVALID: Last 4 characters must be digits")
else:
print(reg_id, "-> VALID")
Run
E-MAIL VALIDATION
"C:\Users\krish\PycharmProjects\Activity Based Learning\Medium\.venv\Scripts\python.exe" "C:\Users\krish\PycharmProjects\Activity Based Learning\Medium\E-MAIL VALIDATION.py"
STUDENT EMAIL VALIDATION
Enter number of students: 1
Enter email address: kavya@gmail.com
kavya@gmail.com -> VALID
REGISTRATION ID VALIDATION
Medium > E-MAIL VALIDATION.py
49:38 CRLF UTF-8 4 spaces Types: Built-in Python 3.14 (Medium)
28°C
Partly sunny
Search
ENG
IN
01:02 PM
27-10-2026